



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 01.08.2026

In addition to part I (General Handout for all courses appended to the Timetable) this portion gives further specific details regarding the course.

Course No : PHA G618
Course Title : Retrosynthetic Analysis
Instructor-in-Charge : Dr. Sandeep Sundriyal
Instructor(s) :
Tutorial/Practical Instructors:

1. Course Description: This course deals with the strategy and tactics involved in the synthesis of organic molecules. The idea is to identify key structural features in the target molecules and then go backward logically to identify appropriate starting materials and chemical reactions for their synthesis. Knowledge and understanding of common organic reagents and reaction mechanisms are required for this course.

2. Scope and Objective of the Course: Most life-saving drugs are organic molecules. This course aims to develop a logical understanding which can be used to propose the synthesis of an organic compound. This course teaches various strategies and tactics together with the common nomenclature used in Retrosynthetic Analysis. At the end of the course, students should be able to propose a synthetic scheme for various drug molecules with appropriate justification.

3. Text Books:

- 1) Organic Synthesis: The disconnection approach by Stuart Warren & Paul Wyatt, Wiley Publishers, IInd edn, 2010. New York.
- 2) Designing Organic Synthesis by Stuart Warren, John Wiley & Sons, IInd edn, 2002. New York. ISBN: 047199612.

4. Reference Books:

- 1) Work book for organic synthesis: The disconnection approach by Stuart Warren & Paul Wyatt, Wiley Publishers, IInd edn, 2010. New York.
- 2) The logic of Chemical Synthesis by E.J. Corey & Xue-Min Chelg, John Wiley & Sons Publishers, Ist edn, 1995. New York. ISBN: 0471115940
- 3) Organic Synthesis by Christine Willis & Martin Wills, Oxford Science Publications, New York. Ist edn, 1996. ISBN: 0198557914

5. Course Plan:

Module No.	Lecture Session	Reference	Learning outcomes
1-2	Introduction & basic concepts of retrosynthetic analysis	TB-1/RB-1: Chap 1-3 RB-2: Chap 1	Glossary and basic principles used in retrosynthesis: synthon, retrons, retrosynthesis, functional group interconversion (FGI), synthetic equivalents and molecular complexity



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3-4	Retrosynthesis of Aromatic compounds	TB-1/RB-1: Chap 2,3	Common reactions on aromatic rings, order of events
5-6	One group C-X disconnections	TB-1/RB-1: chap 4	Disconnecting a bond joining the heteroatom (X) to the rest of the molecule: a C-O, C-N or C-S disconnection
7	Chemoselectivity as strategy	TB-1/RB-1: - Chap 5	Selective reaction on FGs and protection group strategies
8-10	Two group C-X disconnections	TB-1/RB-1: Chap 6	Recognizing two FGs and exploiting the other FG to help discover a better reagent
11	Umpolung strategy	TB-1/RB-1: Chap 7	To exploit reversal of polarity especially in case of carbonyl compounds
12-13	Amine synthesis	TB-1/RB-1: Chap 8	Strategies to introduce amine FG
14-16	One Group and Two Group C-C Disconnections	TB-1/RB-1: Chap 10,13,17	Synthesis of alcohols, aldehydes and ketones; Diels Alder reaction
17	Regioselectivity as strategy	RB-2: Chap 14	how to react one specific part of a single functional group, regioselective alkylation of ketones, conjugate addition
18-20	Alkenes and alkynes	TB-1/RB-1: Chap 15,16	Synthesis of alkenes: elimination, Wittig; exploiting alkynes in FGI
21	Carbonyl condensation as strategy	TB-1/RB-1: Chap 18	Exploiting various condensation reactions: carbonyl FG as nucleophiles and electrophiles
22-33	Two Group C-C Disconnections: 1,3-, 1,5-, 1,2- 1,4-, and 1,6 difunctionalized compounds	TB-1/RB-1: Chap 19-23 and 25, 27, 28	Synthesis of β hydroxy, α,β -unsaturated compounds, conjugate addition, etc.
34	Reconnection as strategy	TB-1/RB-1: Chap 26	Synthesis of 1,2- and 1,4-diCO Compounds by Oxidative C=C Cleavage



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35-38	Synthesis of cyclic molecules	TB-1/RB-1: Chap 29-37, 38	Synthesis of saturated (3, 4, 5 and 6-membered) and heteroaromatic molecules
Remaining hours	Buffer lectures, quizzes, discussion for specific topics, etc.		

6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90 min	30	10/10 AN2	CB
Comprehensive Examination	180 min	35	16/12 AN	CB and/or OB
Continuous assessment (test/assignment)	Continuous (announced or unannounced)	35	As announced by IC	CB and/or OB

7. Chamber Consultation Hour: Every Friday 5.00 – 6.00 pm.

8. Notices: Notices concerning the course will be communicated via email and announced in the class. Hence, students are expected to check their official email on regular basis.

9. Make-up Policy: Generally, make-up is not given as a routine. It is solely dependent on the "genuineness" of the circumstance under which a student fails to appear in a scheduled evaluation component. However, the make-up application should be personally given to Instructor-in-Charge and not slipped into the chamber of the instructor-in-Charge.

10. Note (if any): It is strongly advised that all students prepare their own class notes and relevant information from text and reference material, which would only be allowed for consultation during open book/note assessments. Photocopies of any material, written or printed, will not be permitted.

(Prof. Sandeep Sundriyal)
Instructor-in-charge
Course No. PHA G618